Republic of the Philippines
CAVITE STATE UNIVERSITY
Don Severino delas Alas Campus
Indang, Cavite

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

Name: _____ **Course, Yr. & Sec:** _____
Student Number: _____ **Date:** _____

Instructions:

1. Copy this code to your previous Application.(Practical)

```
import 'package:flutter/material.dart';


void main(){
    runApp(MyApp());
}
```

```
class MyApp extends StatelessWidget {
  const MyApp({super.key});
```

```
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
        debugShowCheckedModeBanner: false,
        title: "My First App",
        theme: ThemeData(colorScheme: ColorScheme.light()),
        home: HomeScreen(),
    );
  }
}
```

```
class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});
```

```
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      // Let the background image go behind the app bar
      extendBodyBehindAppBar: true,
      appBar: AppBar(
        backgroundColor: Colors.transparent,
        elevation: 0,
        foregroundColor: Colors.white,
        title: Row(
          children: [
            ClipOval(
              child: Image.asset(
                "assets/images/logo.png",
                width: 36,
                height: 36,
                fit: BoxFit.cover,
              ),
            ),
            SizedBox(width: 12),
```

Republic of the Philippines
CAVITE STATE UNIVERSITY
Don Severino delas Alas Campus
Indang, Cavite

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

```
          Text("DCIT26", style: TextStyle(fontSize: 16)),
        ],
      ),
    ),
    body: Container(
      width: double.infinity,
      height: double.infinity,
      // Background image with a dark overlay
      decoration: BoxDecoration(
        image: DecorationImage(
          image: AssetImage("assets/images/bakground.jpg"),
          fit: BoxFit.cover,
          colorFilter: ColorFilter.mode(
            Colors.black.withOpacity(0.6),
            BlendMode.darken,
          ),
        ),
      ),
      child: SingleChildScrollView(
        child: Column(
          children: [
            SizedBox(height: 100), // space for the app bar
```

```
            // Search bar
            Padding(
              padding: EdgeInsets.symmetric(horizontal: 30),
              child: TextField(
                decoration: InputDecoration(
                  hintText: "Search here",
                  filled: true,
                  fillColor: Color(0xFFEEEEEE),
                  contentPadding:
                      EdgeInsets.symmetric(horizontal: 20, vertical: 14),
                  border: OutlineInputBorder(
                    borderRadius: BorderRadius.circular(30),
                    borderSide: BorderSide.none,
                  ),
                ),
              ),
            ),
            SizedBox(height: 50),
```

```
            // ---------- Coffee 1 (text left, image right) ----------
            Row(
              children: [
                Expanded(
                  child: Padding(
                    padding: EdgeInsets.only(left: 40, right: 16),
                    child: Column(
```

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

```
                    crossAxisAlignment: CrossAxisAlignment.start,
                    children: [
                      Text(
                        "Coffee 1",
                        style: TextStyle(
                          fontSize: 16,
                          fontWeight: FontWeight.bold,
                          color: Colors.white,
                        ),
                      ),
                      SizedBox(height: 6),
                      Text(
                        "Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Nullam quis varius odio.",
                        style: TextStyle(
                          fontSize: 13,
                          fontWeight: FontWeight.bold,
                          color: Colors.white,
                        ),
                      ),
                      SizedBox(height: 8),
                      ElevatedButton(
                        onPressed: () => print("Coffee 1 Buy Now"),
                        style: ElevatedButton.styleFrom(
                          backgroundColor: Color(0xFFE0E0E0),
                          foregroundColor: Colors.black,
                          shape: RoundedRectangleBorder(
                            borderRadius: BorderRadius.circular(8),
                          ),
                        ),
                        child: Text("Buy Now"),
                      ),
                    ],
                  ),
                ),
              ),
              ClipRRect(
                borderRadius: BorderRadius.only(
                  topLeft: Radius.circular(20),
                  bottomLeft: Radius.circular(20),
                ),
                child: Image.asset(
                  "assets/images/coffee.jpg",
                  width: 160,
                  height: 150,
                  fit: BoxFit.cover,
                ),
              ),
            ],
```

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

```
            ),
            SizedBox(height: 40),
```

```
            // ---------- Coffee 2 (image left, text right) ----------
            Row(
              children: [
                ClipRRect(
                  borderRadius: BorderRadius.only(
                    topRight: Radius.circular(20),
                    bottomRight: Radius.circular(20),
                  ),
                  child: Image.asset(
                    "assets/images/coffee.jpg",
                    width: 160,
                    height: 150,
                    fit: BoxFit.cover,
                  ),
                ),
                Expanded(
                  child: Padding(
                    padding: EdgeInsets.symmetric(horizontal: 30),
                    child: Column(
                      crossAxisAlignment: CrossAxisAlignment.start,
                      children: [
                        Text(
                          "Coffee 2",
                          style: TextStyle(
                            fontSize: 16,
                            fontWeight: FontWeight.bold,
                            color: Colors.white,
                          ),
                        ),
                        SizedBox(height: 6),
                        Text(
                          "Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Nullam quis varius odio.",
                          style: TextStyle(
                            fontSize: 13,
                            fontWeight: FontWeight.bold,
                            color: Colors.white,
                          ),
                        ),
                        SizedBox(height: 8),
                        ElevatedButton(
                          onPressed: () => print("Coffee 2 Buy Now"),
                          style: ElevatedButton.styleFrom(
                            backgroundColor: Color(0xFFE0E0E0),
                            foregroundColor: Colors.black,
                            shape: RoundedRectangleBorder(
```

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

```
                                borderRadius: BorderRadius.circular(8),
                              ),
                            ),
                            child: Text("Buy Now"),
                          ),
                        ],
                      ),
                    ),
                  ),
                ],
              ),
              SizedBox(height: 40),
```

```
              // ---------- Coffee 3 (text left, image right) ----------
              Row(
                children: [
                  Expanded(
                    child: Padding(
                      padding: EdgeInsets.only(left: 40, right: 16),
                      child: Column(
                        crossAxisAlignment: CrossAxisAlignment.start,
                        children: [
                          Text(
                            "Coffee 3",
                            style: TextStyle(
                              fontSize: 16,
                              fontWeight: FontWeight.bold,
                              color: Colors.white,
                            ),
                          ),
                          SizedBox(height: 6),
                          Text(
                            "Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Nullam quis varius odio.",
                            style: TextStyle(
                              fontSize: 13,
                              fontWeight: FontWeight.bold,
                              color: Colors.white,
                            ),
                          ),
                          SizedBox(height: 8),
                          ElevatedButton(
                            onPressed: () => print("Coffee 3 Buy Now"),
                            style: ElevatedButton.styleFrom(
                              backgroundColor: Color(0xFFE0E0E0),
                              foregroundColor: Colors.black,
                              shape: RoundedRectangleBorder(
                                borderRadius: BorderRadius.circular(8),
                              ),
```

Republic of the Philippines
CAVITE STATE UNIVERSITY
Don Severino delas Alas Campus
Indang, Cavite

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

```
                          ),
                          child: Text("Buy Now"),
                        ),
                      ],
                    ),
                  ),
                ),
                ClipRRect(
                  borderRadius: BorderRadius.only(
                    topLeft: Radius.circular(20),
                    bottomLeft: Radius.circular(20),
                  ),
                  child: Image.asset(
                    "assets/images/coffee.jpg",
                    width: 160,
                    height: 150,
                    fit: BoxFit.cover,
                  ),
                ),
              ],
            ),
            SizedBox(height: 40),
          ],
        ),
      ),
    ),
    bottomNavigationBar: BottomNavigationBar(
      type: BottomNavigationBarType.fixed,
      backgroundColor: Colors.white,
      currentIndex: 1,
      selectedItemCoimport 'package:flutter/material.dart';


void main(){
    runApp(MyApp());
}
```

```
class MyApp extends StatelessWidget {
  const MyApp({super.key});
```

```
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
        debugShowCheckedModeBanner: false,
        title: "My First App",
        theme: ThemeData(colorScheme: ColorScheme.light()),
        home: HomeScreen(),
    );
  }
```

```
}
```

```
class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});
```

```
  // The coffee data (const is needed because HomeScreen has a const constructor)
  final List<Map<String, String>> coffees = const [
    {
      "name": "Coffee 1",
      "description":
          "Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam quis varius
odio.",
      "image": "assets/images/coffee.jpg",
    },
    {
      "name": "Coffee 2",
      "description":
          "Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam quis varius
odio.",
      "image": "assets/images/coffee.jpg",
    },
    {
      "name": "Coffee 3",
      "description":
          "Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam quis varius
odio.",
      "image": "assets/images/coffee.jpg",
    },
  ];
```

```
  // One coffee row (image side depends on imageOnRight)
  Widget buildCoffeeItem(Map<String, String> coffee, bool imageOnRight) {
    Widget image = ClipRRect(
      borderRadius: imageOnRight
          ? BorderRadius.only(
              topLeft: Radius.circular(20),
              bottomLeft: Radius.circular(20),
            )
          : BorderRadius.only(
              topRight: Radius.circular(20),
              bottomRight: Radius.circular(20),
            ),
      child: Image.asset(
        coffee["image"]!,
        width: 160,
        height: 150,
        fit: BoxFit.cover,
      ),
    );
```

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

```
    Widget text = Expanded(
      child: Padding(
        padding: EdgeInsets.only(
          left: imageOnRight ? 40 : 30,
          right: imageOnRight ? 16 : 30,
        ),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Text(
              coffee["name"]!,
              style: TextStyle(
                fontSize: 16,
                fontWeight: FontWeight.bold,
                color: Colors.white,
              ),
            ),
            SizedBox(height: 6),
            Text(
              coffee["description"]!,
              style: TextStyle(
                fontSize: 13,
                fontWeight: FontWeight.bold,
                color: Colors.white,
              ),
            ),
            SizedBox(height: 8),
            ElevatedButton(
              onPressed: () => print("${coffee["name"]} Buy Now"),
              style: ElevatedButton.styleFrom(
                backgroundColor: Color(0xFFE0E0E0),
                foregroundColor: Colors.black,
                shape: RoundedRectangleBorder(
                  borderRadius: BorderRadius.circular(8),
                ),
              ),
              child: Text("Buy Now"),
            ),
          ],
        ),
      ),
    );
```

```
    return Row(
      children: imageOnRight ? [text, image] : [image, text],
    );
  }
```

```
  // Search bar + coffee list
```

Republic of the Philippines
CAVITE STATE UNIVERSITY
Don Severino delas Alas Campus
Indang, Cavite

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

```
Widget buildHomePage() {
  return SingleChildScrollView(
    child: Column(
      children: [
        SizedBox(height: 100), // space for the app bar
```

```
        // Search bar (not working yet)
        Padding(
          padding: EdgeInsets.symmetric(horizontal: 30),
          child: TextField(
            decoration: InputDecoration(
              hintText: "Search here",
              filled: true,
              fillColor: Color(0xFFEEEEEE),
              contentPadding:
                  EdgeInsets.symmetric(horizontal: 20, vertical: 14),
              border: OutlineInputBorder(
                borderRadius: BorderRadius.circular(30),
                borderSide: BorderSide.none,
              ),
            ),
          ),
        ),
        SizedBox(height: 50),
```

```
        // Show every coffee, alternating the image side
        for (int i = 0; i < coffees.length; i++) ...[
          buildCoffeeItem(coffees[i], i % 2 == 0),
          SizedBox(height: 40),
        ],
      ],
    ),
  );
}
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    // Let the background image go behind the app bar
    extendBodyBehindAppBar: true,
    appBar: AppBar(
      backgroundColor: Colors.transparent,
      elevation: 0,
      foregroundColor: Colors.white,
      title: Row(
        children: [
          ClipOval(
            child: Image.asset(
              "assets/images/logo.png",
```

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

```
                width: 36,
                height: 36,
                fit: BoxFit.cover,
              ),
            ),
            SizedBox(width: 12),
            Text("DCIT26", style: TextStyle(fontSize: 16)),
          ],
        ),
      ),
      body: Container(
        width: double.infinity,
        height: double.infinity,
        // Background image with a dark overlay
        decoration: BoxDecoration(
          image: DecorationImage(
            image: AssetImage("assets/images/bakground.jpg"),
            fit: BoxFit.cover,
            colorFilter: ColorFilter.mode(
              Colors.black.withOpacity(0.6),
              BlendMode.darken,
            ),
          ),
        ),
        child: buildHomePage(),
      ),
      bottomNavigationBar: BottomNavigationBar(
        type: BottomNavigationBarType.fixed,
        backgroundColor: Colors.white,
        currentIndex: 0,
        selectedItemColor: Colors.orange,
        unselectedItemColor: Colors.black,
        iconSize: 28,
        showSelectedLabels: false,
        showUnselectedLabels: false,
        onTap: (index) => print("Tapped index $index"),
        items: const [
          BottomNavigationBarItem(icon: Icon(Icons.home), label: "Home"),
          BottomNavigationBarItem(icon: Icon(Icons.star), label: "Favorite"),
          BottomNavigationBarItem(icon: Icon(Icons.settings_outlined), label: "Settings"),
          BottomNavigationBarItem(icon: Icon(Icons.share_outlined), label: "Share"),
        ],
      ),
    );
  }
}
lor: Colors.orange,
        unselectedItemColor: Colors.black,
        iconSize: 28,
```

DCIT 26: Application Development and Emerging Technologies
Laboratory Exercise 4: StateFulWidget
Prepared by: John Aimiel C. Pena

```
        showSelectedLabels: false,
        showUnselectedLabels: false,
        onTap: (index) => print("Tapped index $index"),
        items: const [
          BottomNavigationBarItem(icon: Icon(Icons.home), label: "Home"),
          BottomNavigationBarItem(icon: Icon(Icons.star), label: "Favorite"),
          BottomNavigationBarItem(icon: Icon(Icons.settings_outlined), label: "Settings"),
          BottomNavigationBarItem(icon: Icon(Icons.share_outlined), label: "Share"),
        ],
      ),
    );
  }
}
```

2. Make the Search bar and Navigation Bar Work

Laboratory Grading Rubric			
Criteria	**Definition**	**Points Possible**	**Earned Points**
Base Code Setup	Uses the provided base code as the starting point, with the assets/images/ folder and pubspec.yaml set up correctly so the coffee screen loads without asset errors.	15	
Bottom Navigation Functionality	Tapping each tab shows the correct page (Home, Favorites, Settings, Share), and the selected icon is highlighted. The background and app bar stay in place while the page changes.	20	
Search Bar Functionality	Filters the coffee list as the user types, ignoring uppercase and lowercase, and shows “No coffee found” when nothing matches. The typed text stays when switching tabs.	20	
Data-Driven Coffee List & Layout	Displays the coffees from the coffees list using a loop and buildCoffeeItem, with the image side alternating correctly, including after the list is filtered.	15	
State Management	Converts HomeScreen to a StatefulWidget with state variables for the selected tab and search text, updates them with setState, and uses a TextEditingController that is released in dispose().	10	
Application Functionality & Presentation	The app runs successfully, and navigation and search work together without errors. The coffee screen keeps its original design, with no overflow or broken elements.	15	
Organization & Code Quality	Files are properly organized and named; code is readable, consistent, and free from unnecessary errors.	5	
Total		**100**	